Experiment no:-01

Title: Study and Simulation of the 3D Radiation Pattern of a Dipole Antenna

Objective:

To study and visualize the three-dimensional (3D) radiation pattern of a half-wave dipole antenna using numerical simulation in Python.

Theory:

A dipole antenna is a fundamental antenna type that radiates electromagnetic energy primarily in the plane perpendicular to its axis. The radiation intensity is maximum in directions perpendicular to the dipole and zero along its axis.

For a half-wave dipole, the radiation pattern is given by:

$$R(\theta) = |\sin(\theta)|$$

where θ is the polar angle measured from the antenna axis. The pattern is rotationally symmetric around the axis, forming a characteristic "donut-shaped" 3D radiation pattern.

The Cartesian coordinates for plotting the 3D radiation pattern are derived as:

$$X = R\sin(\theta)\cos(\phi)$$
$$Y = R\sin(\theta)\sin(\phi)$$
$$Z = R\cos(\theta)$$

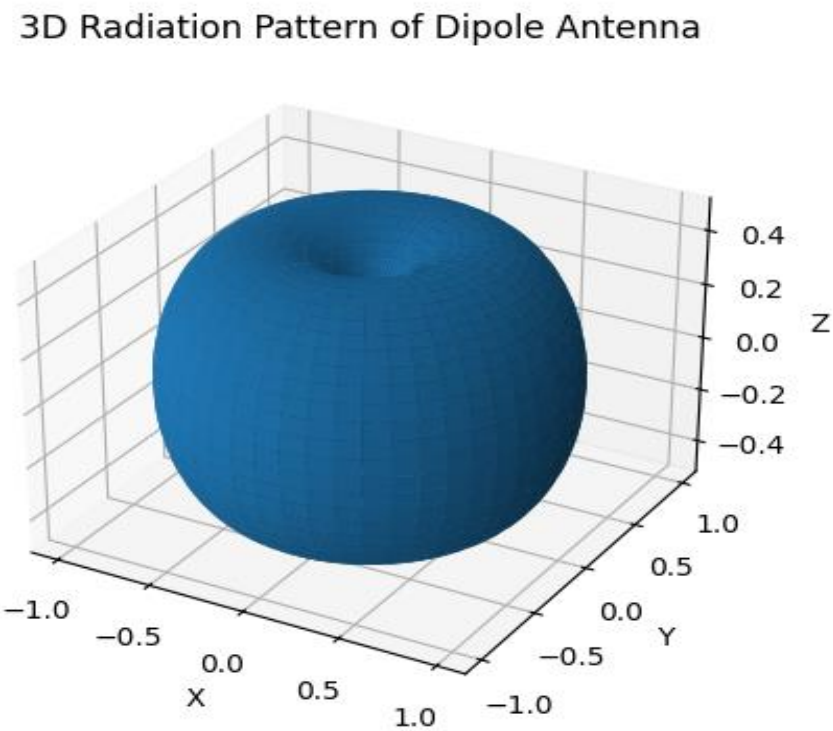
where ϕ is the azimuthal angle.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt
theta = np.linspace(0, np.pi, 100)
phi = np.linspace(0, 2*np.pi, 100)
theta, phi = np.meshgrid(theta, phi)
R = np.abs(np.sin(theta))
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)
fig = plt.figure()
ax = fig.add_subplot(projection='3d')
```

```
ax.plot_surface(X, Y, Z)
ax.set_title("3D Radiation Pattern of Dipole Antenna")
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")
plt.show()
```

output:



Result and Discussion:

- The simulation successfully visualized the 3D radiation pattern of a half-wave dipole antenna.
- The maximum radiation occurs in the plane perpendicular to the antenna axis, while the radiation along the axis is zero, as expected theoretically.
- This 3D “donut-shaped” pattern demonstrates the characteristic directional properties of a dipole antenna.
- Such visualization aids in understanding antenna behavior and is useful for antenna design and placement in practical applications.

Experiment no:-02

Title: Study and Simulation of the 3D Radiation Pattern of an Isotropic Antenna

Objective:

- To understand the radiation characteristics of an isotropic antenna.
- To simulate and visualize the 3D radiation pattern using Python.
- To verify that an isotropic antenna radiates equally in all directions.

Theory:

An isotropic antenna is an ideal and hypothetical antenna that radiates electromagnetic energy equally in all directions in space. It is used as a reference antenna to compare the performance of practical antennas.
The radiation intensity of an isotropic antenna is constant for all angles, meaning it does not depend on the angular coordinates (θ, φ).
Mathematically:

$$R(\theta, \phi) = constant$$

In spherical coordinates:

- θ (theta) varies from 0 to π
- φ (phi) varies from 0 to 2π

When converted into Cartesian coordinates:

$$X = R\sin\ \theta\cos\ \phi$$
$$Y = R\sin\ \theta\sin\ \phi$$
$$Z = R\cos\ \theta$$

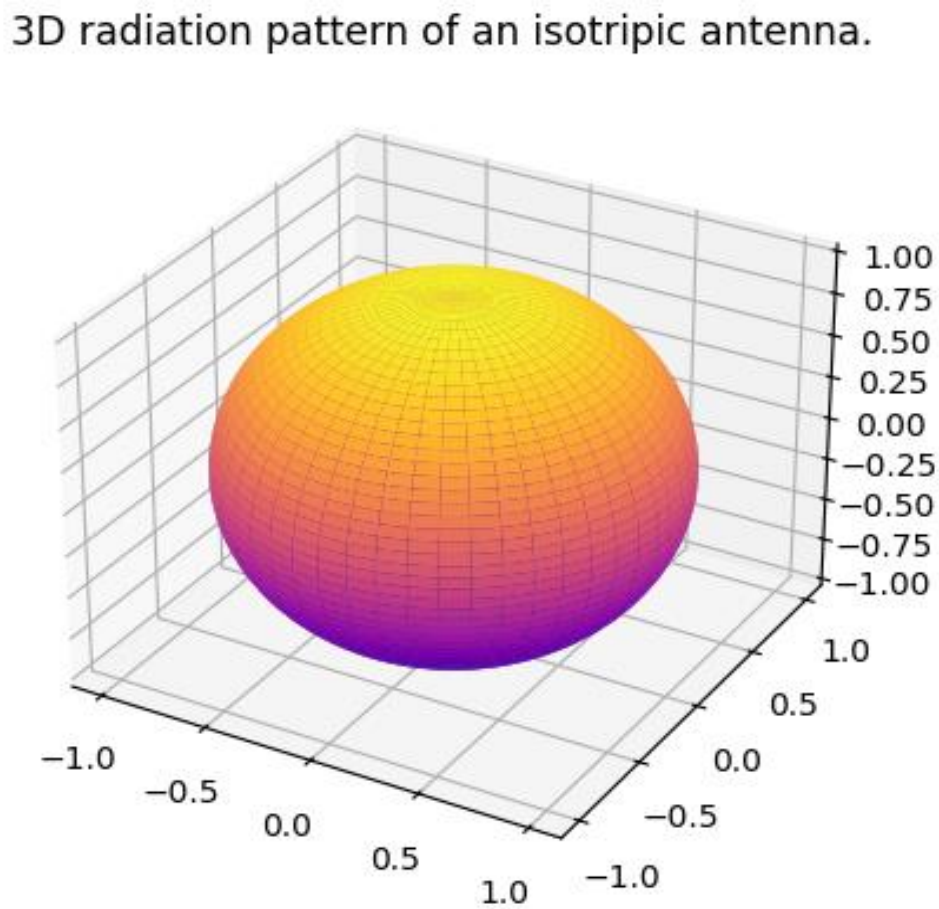
Since $R = 1$(constant), the radiation pattern forms a perfect sphere, representing uniform radiation in all directions.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt
theta = np.linspace(0, np.pi, 200)
phi = np.linspace(0, 2*np.pi, 200)
theta, phi = np.meshgrid(theta, phi)
R = np.ones_like(theta)
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)
```

```
ax = plt.figure().add_subplot(projection='3d')
ax.plot_surface(X, Y, Z, cmap="plasma")
ax.set_title("Isotropic Antenna (3D)")
plt.show()
```

Output:



Result and Discussion:

The simulation successfully produced a spherical 3D radiation pattern, confirming that the isotropic antenna radiates uniformly in all directions. Unlike practical antennas, which have directional patterns, the isotropic antenna serves as an ideal reference model.

This experiment demonstrates that:

- The 3D plot forms a perfect sphere, validating theoretical expectations.
- Isotropic antennas are not physically realizable but are essential for antenna analysis and gain comparison.

Experiment no:-03

Title: Study and Simulation of the 3D Radiation Pattern of a Horn Antenna

Objective:

- To study the radiation characteristics of a horn antenna.
- To simulate and visualize the 3D radiation pattern using Python.
- To understand how antenna directivity affects radiation distribution in space.

Theory:

A horn antenna is a widely used directive antenna that operates efficiently at microwave frequencies. It is formed by flaring out a waveguide, which improves impedance matching and directs electromagnetic waves into free space.

The radiation pattern of a horn antenna is directional, meaning most of the power is radiated in a specific direction (main lobe), with minimal radiation in other directions (side lobes).

In this simulation, the radiation intensity is approximated using:

$$R(\theta) = |\cos(\theta)|^n$$

Where:

- θ= elevation angle
- n= directivity factor (higher $n \rightarrow$ narrower beam)

The 3D radiation pattern is obtained by converting spherical coordinates into Cartesian coordinates:

$$X = R\sin\theta\cos\phi$$
$$Y = R\sin\theta\sin\phi$$
$$Z = R\cos\theta$$

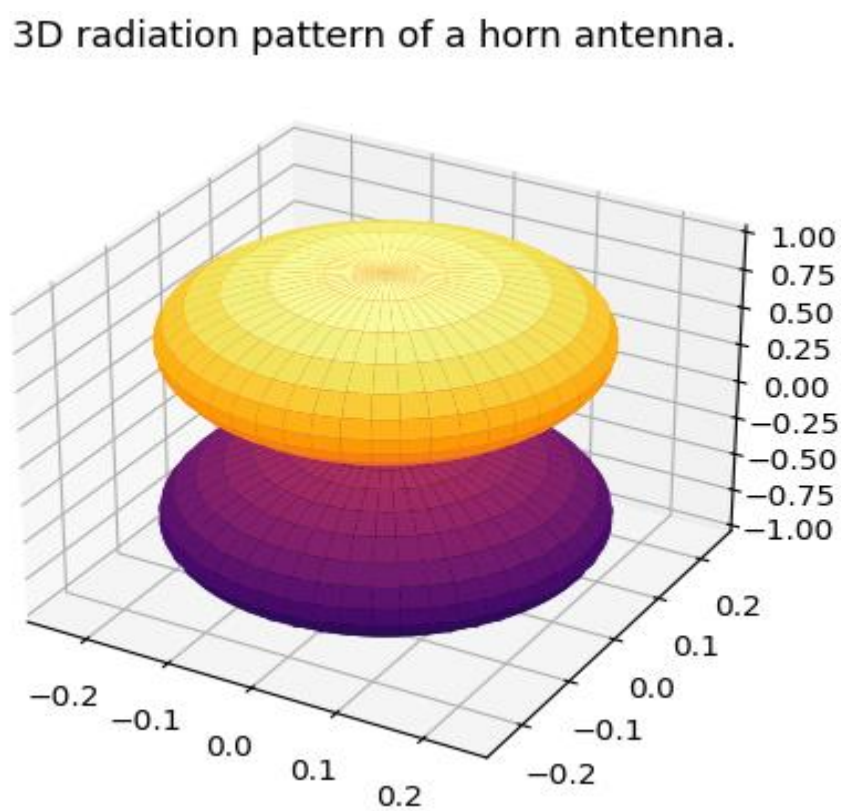
This produces a three-dimensional visualization of how the antenna radiates energy in space.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt
theta = np.linspace(0, np.pi, 250)
phi = np.linspace(0, 2*np.pi, 250)
theta, phi = np.meshgrid(theta, phi)
n = 6
```

```
R = np.abs(np.cos(theta))**n
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)
ax = plt.figure().add_subplot(projection='3d')
ax.plot_surface(X, Y, Z, cmap="inferno")
ax.set_title("3D radiation pattern of a horn antenna.")
plt.show()
```

Output:



Result and Discussion:

- The horn antenna exhibits a highly directional radiation pattern, concentrating energy in one primary direction.
- Increasing the value of $n$results in a narrower and more focused beam, indicating higher directivity.
- The 3D plot confirms that radiation intensity is maximum along the axis and decreases rapidly away from it.

Experiment no:-04

Title: Study and Simulation of the 2D Radiation Pattern of a Linear Antenna Array

Objective:

- To study the radiation characteristics of a linear antenna array.
- To simulate and visualize the 2D radiation pattern using computational tools.
- To understand the effect of array parameters such as number of elements and spacing on the radiation pattern.

Theory:

A linear antenna array consists of multiple antenna elements arranged in a straight line with equal spacing. The overall radiation pattern of the array depends on the constructive and destructive interference of waves radiated by individual elements.

The radiation pattern of a linear array is described by the **Array Factor (AF)**, given by:

$$AF = \frac{\sin\left(\frac{Nkd\cos\,\theta}{2}\right)}{N\sin\left(\frac{kd\cos\,\theta}{2}\right)}$$

Where:

- N= Number of elements
- d= spacing between elements in wavelength
- $k = 2\pi/\lambda$= Wave number
- θ= Angle of observation

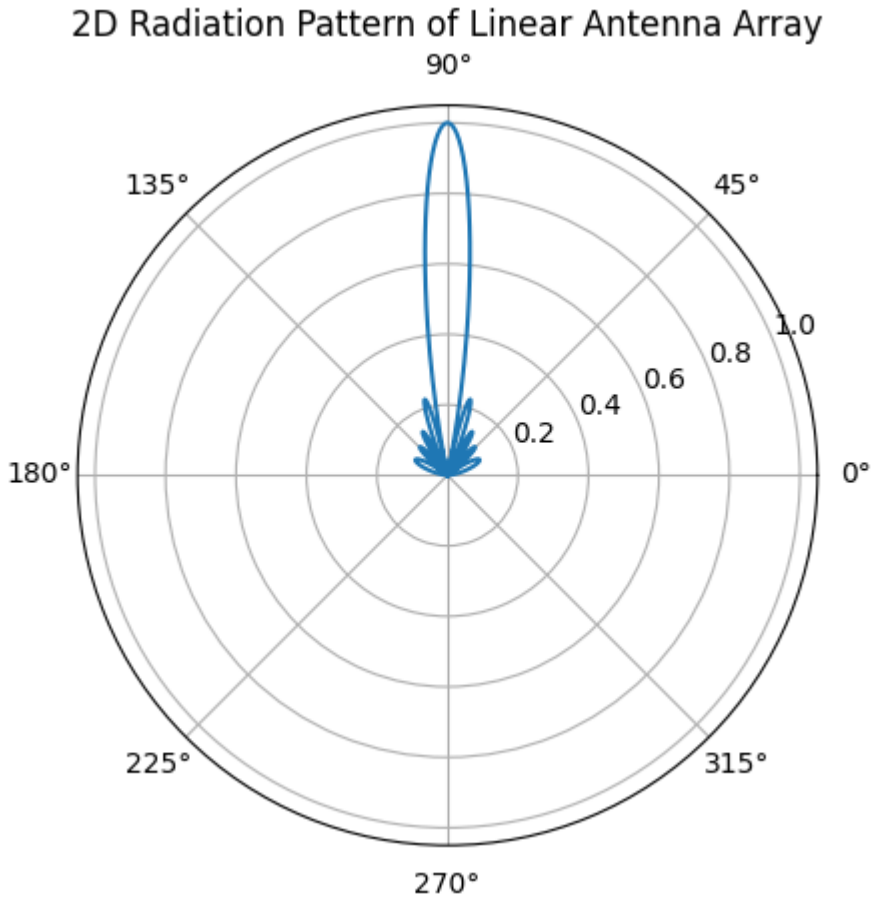
The total radiation pattern is proportional to the magnitude of the array factor. The pattern shows a **main lobe** in the direction of maximum radiation and **side lobes** due to interference.

When element spacing is $d = \lambda/2$, it helps to avoid grating lobes and provides a well-defined directional pattern.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi, 800)
N = 10
d = 0.5
k = 2*np.pi
num = np.sin(N * k * d * np.cos(theta) / 2)
den = N * np.sin(k * d * np.cos(theta) / 2)
den = np.where(den == 0, 1e-10, den)
AF = np.abs(num / den)

ax = plt.subplot(111, projection="polar")
ax.plot(theta, AF)
ax.set_title("2D Radiation Pattern of Linear Antenna Array")
plt.show()
```

Output:



Result and Discussion:

From the simulation, it is observed that the linear antenna array produces a **directional radiation pattern**. The main lobe indicates the direction of maximum radiation, while the side lobes represent smaller unwanted radiations.
Increasing the number of elements ($N = 10$) results in a **narrower main lobe**, which improves directivity. The chosen spacing $d = 0.5\lambda$ensures that no grating lobes are formed and the radiation pattern remains efficient.
The use of numerical simulation helps visualize how array parameters influence antenna behavior. Overall, the experiment successfully demonstrates the radiation characteristics of a linear antenna array in 2D.

Experiment no:-05

Title: Study and Simulation of the 2D Radiation Pattern of a Broadside 3-Element Array Antenna

Objective:

- To study the radiation characteristics of a broadside linear array antenna.
- To simulate and visualize the 2D radiation pattern for a 3-element array using Python.
- To understand how element spacing and phase affect the array pattern.

Theory:

A **broadside array antenna** is a linear arrangement of identical radiating elements where the maximum radiation is directed perpendicular (broadside) to the axis of the array.

For a uniform linear array with **N elements**, equal amplitude, and zero phase difference, the **Array Factor (AF)** is given by:

$$AF = \sum_{n=0}^{N-1} e^{jnkd\cos\theta}$$

Where:

- N= number of elements
- $k = \frac{2\pi}{\lambda}$= wave number
- d= distance between elements
- θ= observation angle

For a **broadside array**:

- Phase difference = 0
- Maximum radiation occurs at $\theta = 90°$

In this experiment:

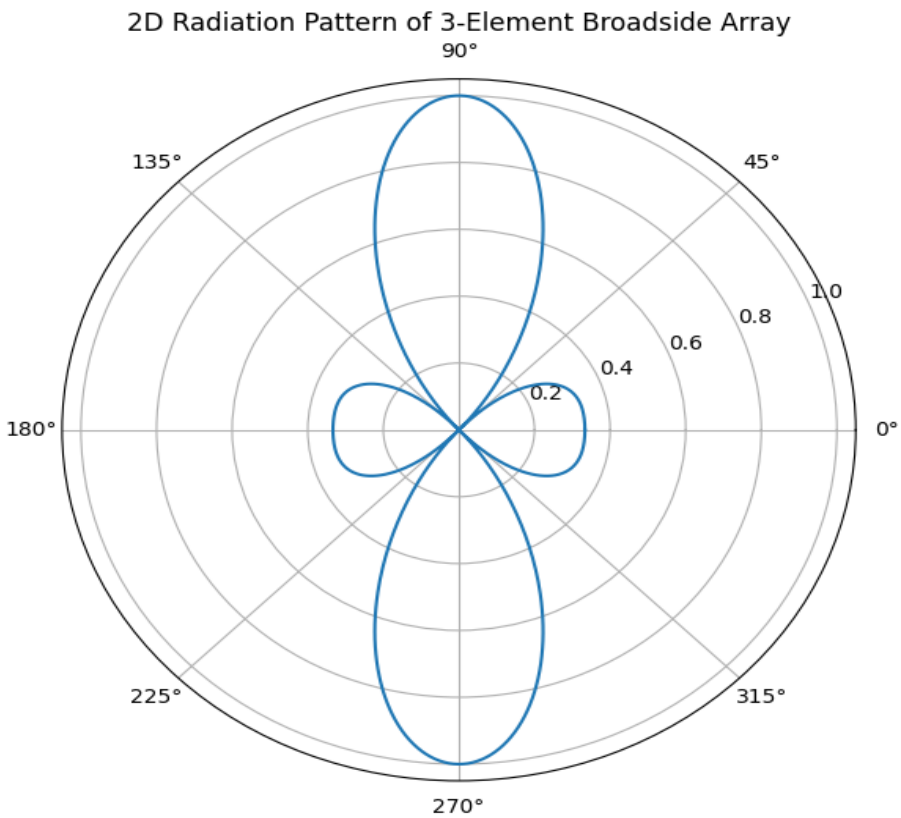
- Number of elements, $N = 3$
- Element spacing, $d = \lambda/2$

The radiation pattern is obtained by calculating the magnitude of the array factor and normalizing it.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt

lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 3
theta = np.linspace(0, 2*np.pi, 1000)
AF = np.zeros_like(theta, dtype=complex)

for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)
ax.set_title("2D Radiation Pattern of 3-Element Broadside Array")
plt.show()
```

Output:



Result and Discussion:

The simulation successfully demonstrates the radiation pattern of a 3-element broadside array antenna. The main observations are:

- The **maximum radiation occurs at broadside direction (θ = 90°)**, which validates the theoretical expectation.
- The presence of **minor side lobes** is observed, which is a natural effect of having multiple elements.
- Using spacing $d = \lambda/2$prevents the formation of unwanted **grating lobes**, ensuring efficient radiation.
- Increasing the number of elements would result in a **narrower main lobe** and improved directivity.

Overall, the experiment confirms that a broadside array antenna provides directional radiation with improved gain compared to a single element antenna.

Experiment no:-06

Title: Study and Plot of the Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Phase (d = λ/2)

Objective:

- To study the radiation characteristics of two isotropic point sources.
- To plot the far-field radiation pattern when both sources have equal amplitude and are in phase.
- To analyze the effect of spacing (d = λ/2) on the radiation pattern.

Theory:

An isotropic radiator is an ideal antenna that radiates equally in all directions. When two such sources are placed at a distance **d** apart and excited with equal amplitude and phase, their radiation fields combine through the principle of superposition.

The total radiation pattern is determined by the **Array Factor (AF)**, given by:

$$AF = 2\cos\left(\frac{kd}{2}\cos\theta\right)$$

Where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- d is the spacing between the sources
- θ is the observation angle

For **d = λ/2**, the equation becomes:

$$AF = 2\cos\left(\frac{\pi}{2}\cos\theta\right)$$

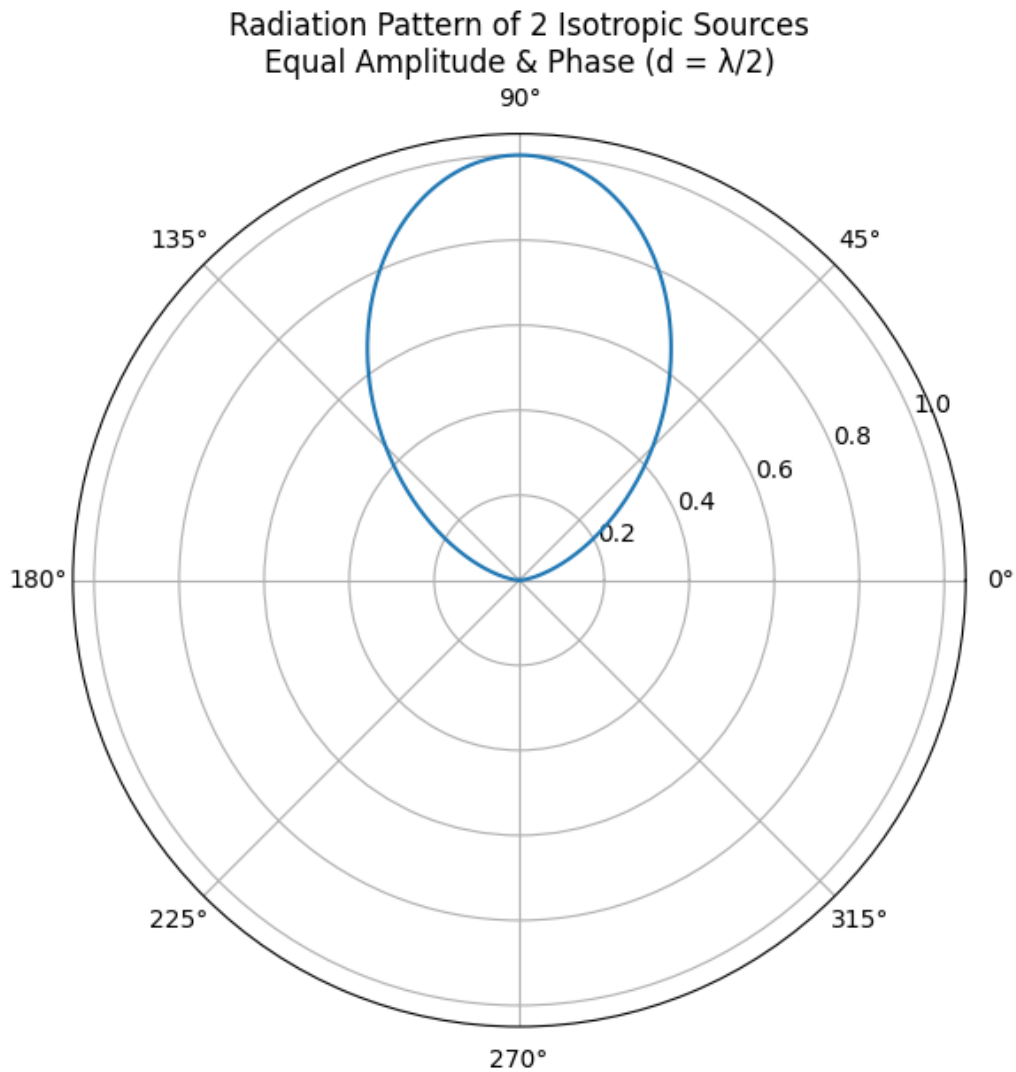
The pattern is then normalized to obtain a clear graphical representation. The radiation pattern shows how power is distributed in space.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam
theta = np.linspace(0, np.pi, 1000)
```

```
AF = 2 * np.cos((k * d / 2) * np.cos(theta))
AF_norm = np.abs(AF) / np.max(np.abs(AF))

plt.figure(figsize=(6,6))
plt.polar(theta, AF_norm)
plt.title("Radiation Pattern of 2 Isotropic Sources\nEqual Amplitude &
Phase (d = λ/2)")
plt.show()
```

Output:



Result and Discussion:

The radiation pattern of two isotropic point sources with equal amplitude and phase and spacing **$d = \lambda/2$** is successfully obtained. The pattern is **bidirectional**, with two main lobes in opposite directions.
It is observed that:

- Constructive interference occurs at $\theta = 90°$, producing maximum radiation.
- Destructive interference occurs at $\theta = 0°$ and $180°$, producing nulls.
- The spacing of $\lambda/2$ ensures a simple pattern without grating lobes.

Thus, the experiment confirms that antenna spacing and phase significantly influence the radiation characteristics. This configuration is commonly used in antenna arrays to achieve directional radiation.

Experiment no:-07

Title: Study and Plot of Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Opposite Phase (d = λ/2)

Objective:

To study and visualize the far-field radiation pattern of two isotropic point sources separated by a distance of λ/2, having equal amplitude and opposite phase, and to analyze the resulting interference pattern.

Theory:

An isotropic source radiates energy equally in all directions. When two such sources are placed at a distance $d = \frac{\lambda}{2}$, their combined radiation pattern depends on their relative phase and spacing.

For two sources with:

- Equal amplitude
- Opposite phase difference (β = π)

The array factor (AF) is given by:

$$AF = 2\sin\left(\frac{kd}{2}\cos\theta\right)$$

where,

$k = \frac{2\pi}{\lambda}$ is the wave number

θ is the observation angle

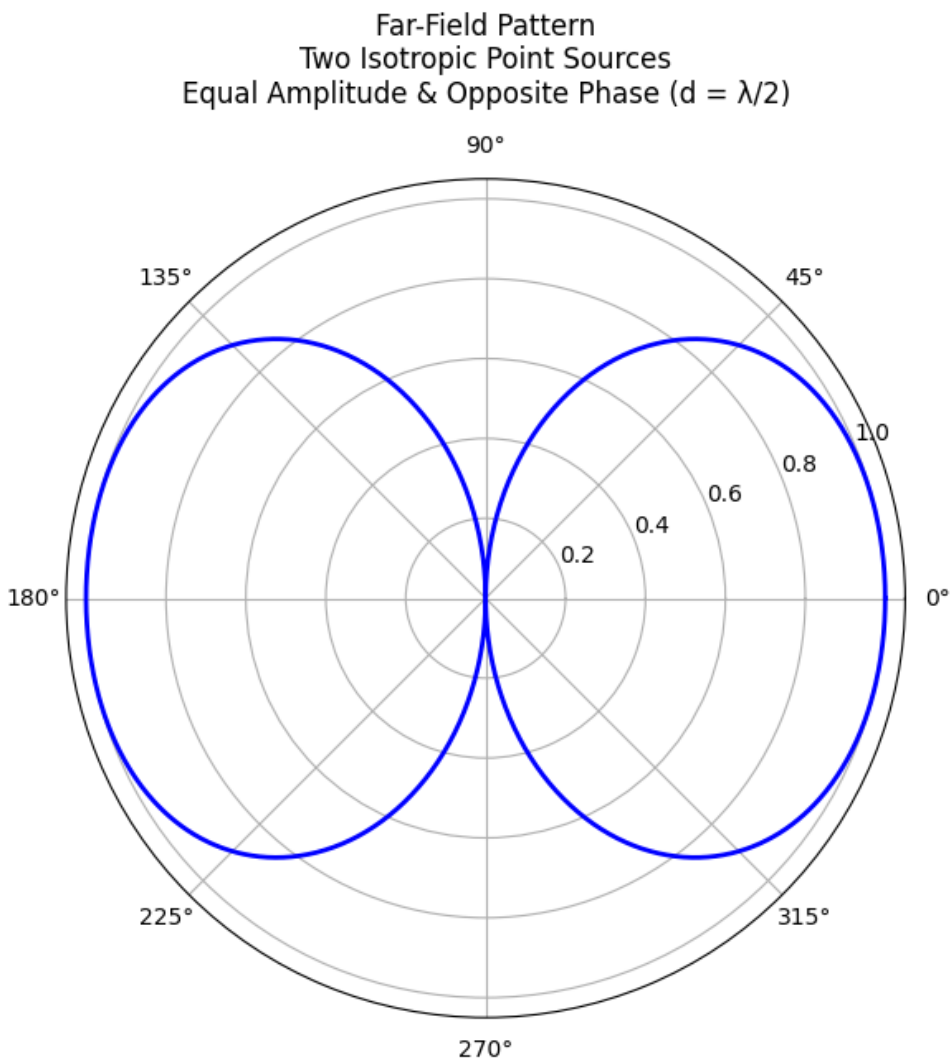
Since the sources are out of phase, destructive interference occurs in certain directions, producing nulls, while constructive interference produces maxima. The resulting radiation pattern is directional, unlike a single isotropic source.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam
theta = np.linspace(0, 2*np.pi, 1000)
AF = 2 * np.sin((k * d / 2) * np.cos(theta))
```

```
AF_norm = np.abs(AF) / np.max(np.abs(AF))

fig = plt.figure(figsize=(7, 7))
ax = fig.add_subplot(111, projection='polar')
ax.plot(theta, AF_norm, color='blue', linewidth=2)
ax.set_title("Far-Field Pattern\nTwo Isotropic Point Sources\nEqual
Amplitude & Opposite Phase (d = λ/2)",
            va='bottom', pad=30, fontsize=12)
ax.grid(True)
plt.tight_layout()
plt.show()
```

Output:



Result and Discussion:

The plotted far-field radiation pattern shows a figure-eight (bidirectional) shape.
Key observations:

- Maximum radiation occurs at $\theta = 90°$and $270°$, where the waves from both sources add constructively.
- Nulls (zero radiation) occur at $\theta = 0°$and $180°$, due to destructive interference caused by the opposite phase.
- The pattern is symmetrical about the axis perpendicular to the line joining the sources.
- The radiation is strongest in directions broadside to the array and weakest along the array axis.

The experiment confirms that two isotropic sources with equal amplitude and opposite phase produce a directional radiation pattern with clear maxima and nulls. The spacing $d = \frac{\lambda}{2}$plays a crucial role in shaping the interference pattern, making the system useful for antenna array design.

Experiment no:-08

Title: Study and Plot of Far-Field Radiation Pattern of Two Isotropic Point Sources with Unequal Amplitude and Opposite Phase (d = λ/2)

Objective:

- To study the far-field radiation pattern of two isotropic point sources.
- To analyze the effect of unequal amplitudes and opposite phase on the radiation pattern.
- To plot the normalized array factor using Python.

Theory:

In antenna array systems, the overall radiation pattern depends on the **array factor (AF)**, which is determined by the relative amplitudes, phases, and spacing between elements.

For two isotropic point sources separated by distance $d = \frac{\lambda}{2}$, with unequal amplitudes $A_1$and A_2, and opposite phase, the array factor is:

$$AF = A_1 e^{j\psi} - A_2 e^{-j\psi}$$

where,

$$\psi = \frac{kd}{2}\cos\ \theta \text{and} k = \frac{2\pi}{\lambda}$$

- θ= observation angle
- λ= wavelength
- k= wave number

Because the sources are in **opposite phase**, a subtraction occurs in the array factor.

When amplitudes are unequal ($A_1 \neq A_2$), the radiation pattern becomes **asymmetric**, unlike the symmetric pattern seen in equal amplitude cases.

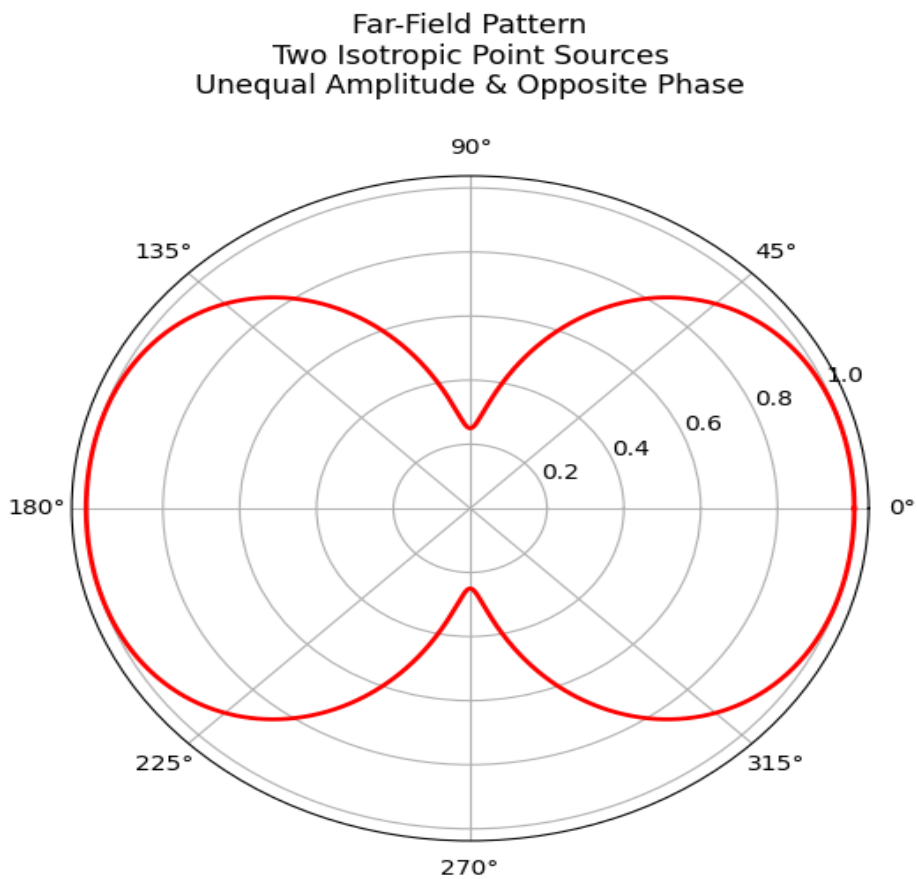
The normalized array factor is given by:

$$AF_{norm} = \frac{|\ AF\ |}{\max\ (|\ AF\ |)}$$

Source code:

```
import numpy as np
import matplotlib.pyplot as plt

lam = 1.0
d = lam / 2
k = 2 * np.pi / lam
A1 = 1.0
A2 = 0.6

theta = np.linspace(0, 2*np.pi, 1000)
psi = (k * d / 2) * np.cos(theta)
AF = A1 * np.exp(1j * psi) - A2 * np.exp(-1j * psi)
AF_norm = np.abs(AF) / np.max(np.abs(AF))

fig = plt.figure(figsize=(7, 8)) # Fig size ektu boro kora hoyeche
ax = fig.add_subplot(111, polar=True)
ax.plot(theta, AF_norm, color='r', linewidth=2)
plt.title("Far-Field Pattern\nTwo Isotropic Point Sources\nUnequal
Amplitude & Opposite Phase",
        fontsize=12, pad=40)
plt.subplots_adjust(top=0.80)
plt.show()
```

Output:



Result and Discussion:

- The radiation pattern of two isotropic sources with unequal amplitudes and opposite phase is asymmetric.
- The main lobe shifts toward the direction of the stronger source ($A_1 = 1.0$).
- The weaker source ($A_2 = 0.6$) reduces radiation intensity in the opposite direction.
- Unlike equal amplitude cases, perfect nulls may not occur, and sidelobes become uneven.
- The phase difference (180°) creates destructive interference, while unequal amplitudes prevent complete cancellation.

Thus, it is concluded that both amplitude imbalance and phase difference significantly affect the shape and direction of the radiation pattern.

Experiment no:-09

Title: Study and Simulation of the 2D Radiation Pattern of a Broadside 5-Element Array Antenna

Objective:

- To study the radiation characteristics of a broadside linear array antenna.
- To simulate and plot the 2D radiation pattern for a 5-element array.
- To understand the effect of element spacing and number of elements on the radiation pattern.

Theory:

A broadside array antenna is a linear array in which the maximum radiation is perpendicular (broadside direction) to the axis of the array. For a uniform linear array with N elements, equal spacing d, and equal amplitude excitation, the Array Factor (AF) is given by:

$$AF = \sum_{n=0}^{N-1} e^{jnkd\cos\theta}$$

Where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- d is the spacing between elements
- θ is the observation angle

For a broadside array, the phase difference between elements is zero, and maximum radiation occurs at:

$$\theta = 90^\circ$$

In this experiment:

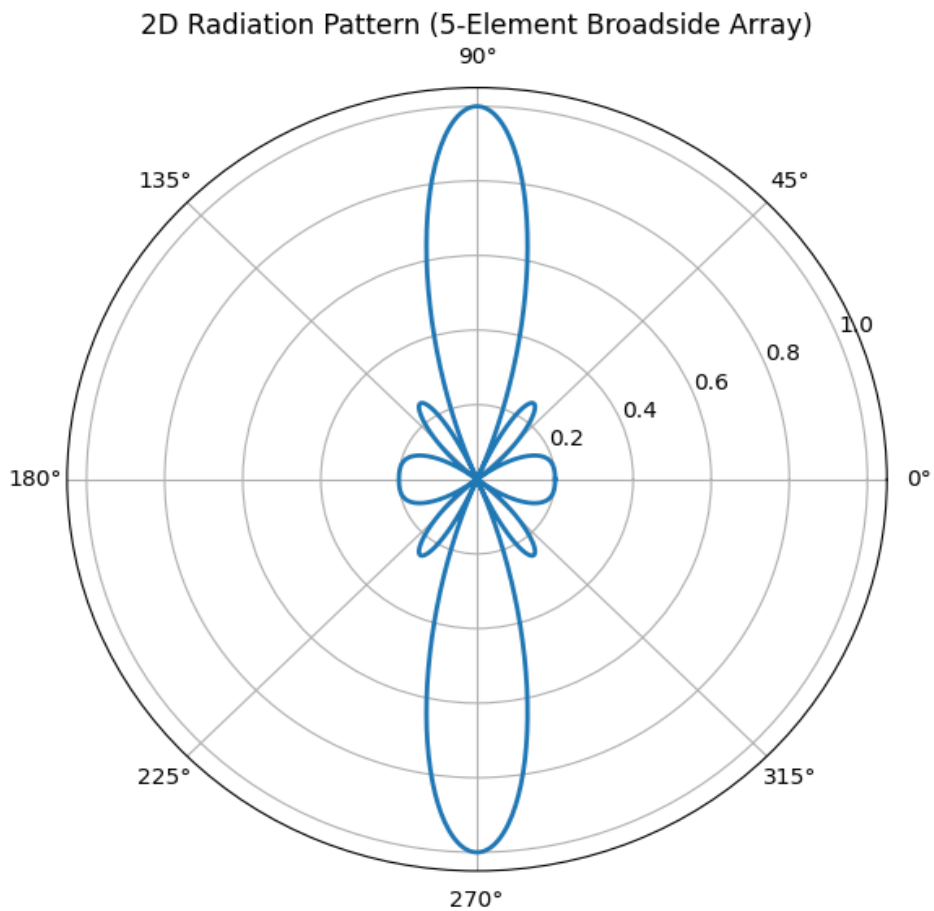
- Number of elements, $N = 5$
- Element spacing, $d = \frac{\lambda}{2}$

The radiation pattern is obtained by calculating and normalizing the magnitude of the array factor.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt

lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 5
theta = np.linspace(0, 2*np.pi, 1000)
n_values = np.arange(-(N-1)/2, (N-1)/2 + 1)
AF = np.zeros_like(theta, dtype=complex)
for n in n_values:
    AF += np.exp(1j * n * k * d * np.cos(theta))
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag, linewidth=2)
ax.set_title("2D Radiation Pattern (5-Element Broadside Array)")
ax.grid(True)
plt.show()
```

Output:



Result and Discussion:

- The simulation successfully demonstrates the radiation pattern of a **5-element broadside array antenna**.
- The **main lobe** is directed perpendicular to the array axis, confirming broadside behavior.
- The radiation pattern is **symmetrical**, indicating uniform excitation and proper element positioning.
- **Side lobes** are present due to multiple elements, but their amplitude is lower than the main lobe.
- The use of $d = \lambda/2$ensures no grating lobes appear, resulting in a clean radiation pattern.
- Increasing the number of elements would further **narrow the main lobe** and improve directivity.

Experiment no:-10

Title: Study and Simulation of the 2D Radiation Pattern of a Broadside 7-Element Array Antenna

Objective:

- To study the radiation characteristics of a 7-element broadside array antenna.
- To simulate and visualize the 2D radiation pattern using Python.
- To analyze the directivity and beam formation of the antenna array.

Theory:

A broadside array antenna is a linear arrangement of identical radiating elements where the main radiation is perpendicular to the axis of the array.

For a uniform linear array with N elements, the total radiation pattern is determined by the Array Factor (AF):

$$AF = \sum_{n=-(N/2)}^{N/2} e^{jnkd\cos\theta}$$

Where:

- N= Number of elements
- d= Spacing between elements
- λ= Wavelength
- $k = \frac{2\pi}{\lambda}$= Wave number
- θ= Angle of observation

For a broadside array:

- The phase difference between elements is zero.
- Maximum radiation occurs at $\theta = 90°$.

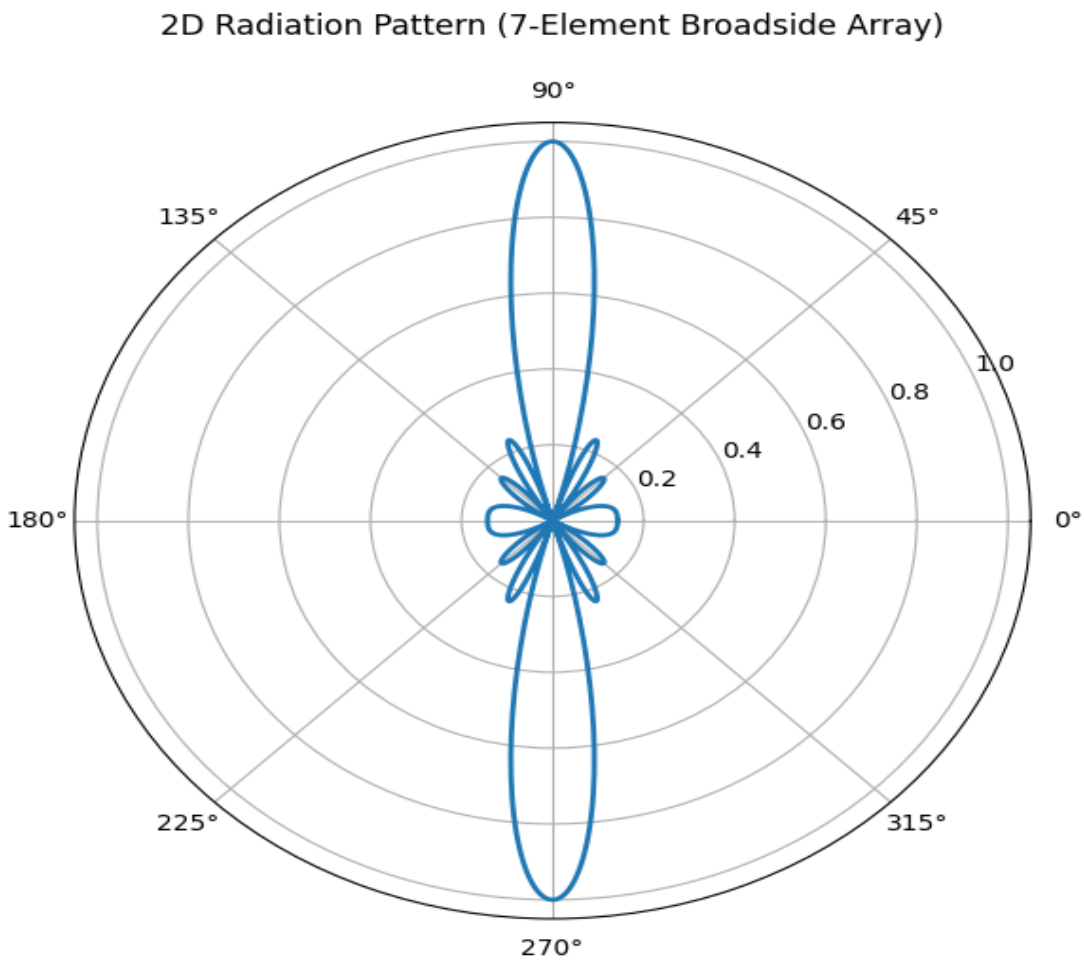
In this experiment:

- Number of elements $N = 7$
- Element spacing $d = \lambda/2$
- Symmetrical placement of elements ensures a centered radiation pattern.

Source code:

```
import numpy as np
import matplotlib.pyplot as plt

lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 7

theta = np.linspace(0, 2*np.pi, 1000)
AF = np.zeros_like(theta, dtype=complex)

for n in range(-(N//2), (N//2)+1):
    AF += np.exp(1j * n * k * d * np.cos(theta))
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag, linewidth=2)
ax.set_title("2D Radiation Pattern (7-Element Broadside Array)\n")
ax.grid(True)
plt.show()
```

Output:



Result and Discussion

The simulation of the 7-element broadside array antenna shows that:

- The maximum radiation occurs at $\theta = 90°$, confirming broadside behavior.
- Increasing the number of elements (N = 7) results in a narrower main lobe, which indicates higher directivity.
- Side lobes are present but significantly smaller than the main lobe.
- The use of $d = \lambda/2$prevents grating lobes and ensures proper radiation characteristics.

Overall, the simulation successfully demonstrates how a multi-element array improves antenna performance by enhancing directivity and controlling radiation patterns.